



Lunar Resource Exercise Design Review

Group 2: Nina M, Ilya Y, Fabio G, August L, Till L, Henning W

Concept of Operations (ConOps)

1. Scan „first-row-cubes“

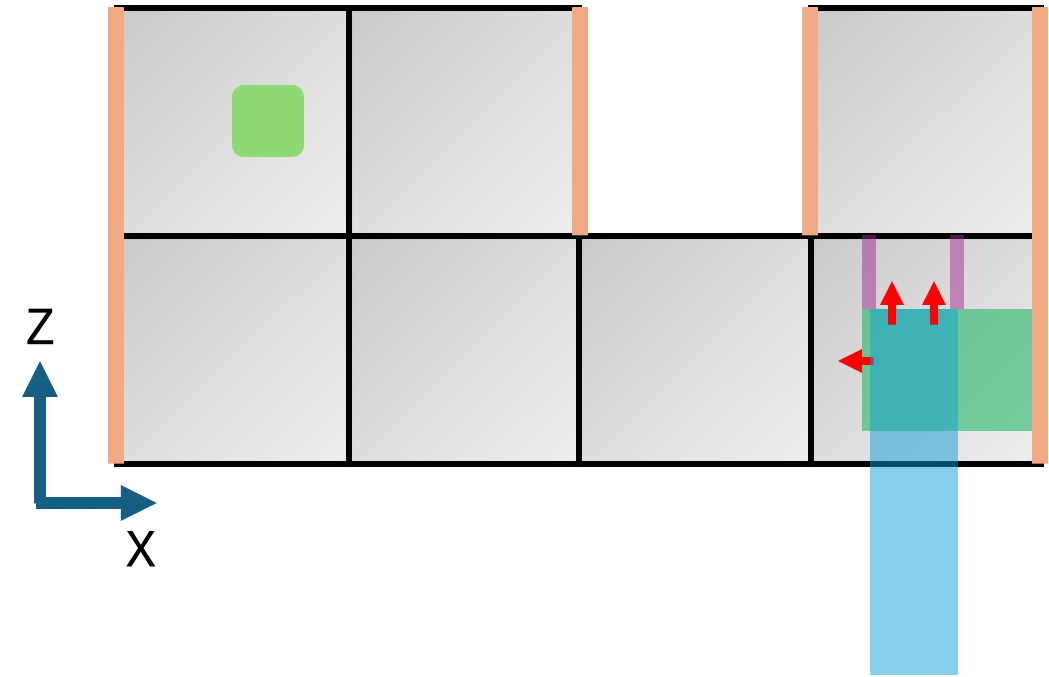
(= cubes which are not partially or fully covered by other cubes from the sensors perspective).

1.1 Turn on the system.

1.2 Save the $x=0$ position

1.3 Drive the System along the X axis using the X-Motor, scan for Cubes using the distance sensor and store their positions.

1.4 Utilising the bump sensor, save the x_{max} position for calibration ($T=20$ s).



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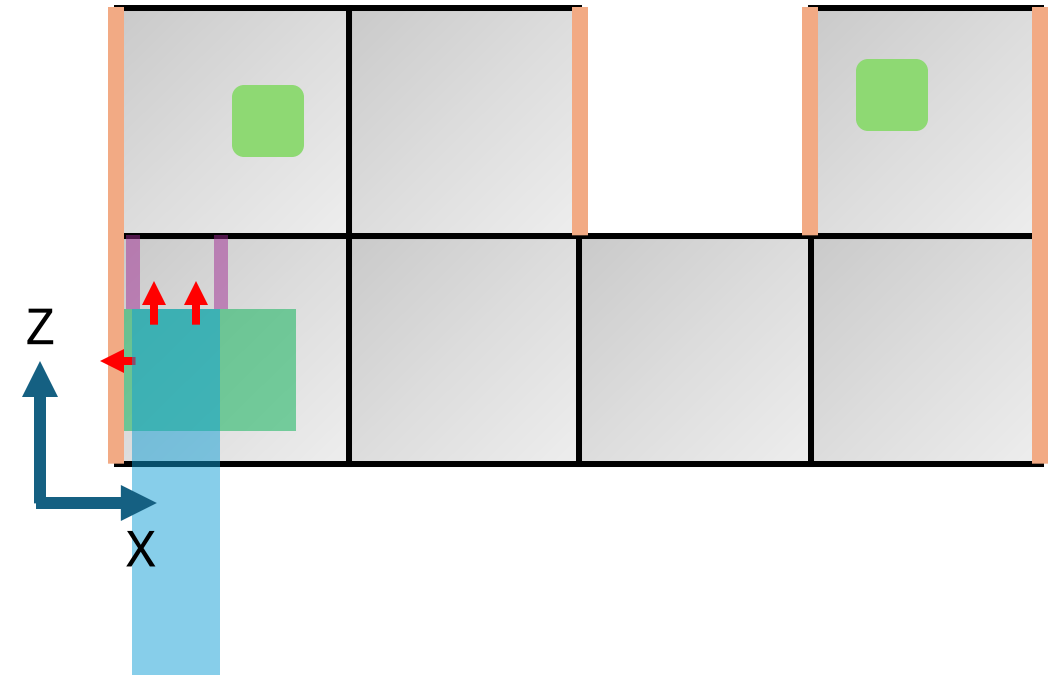
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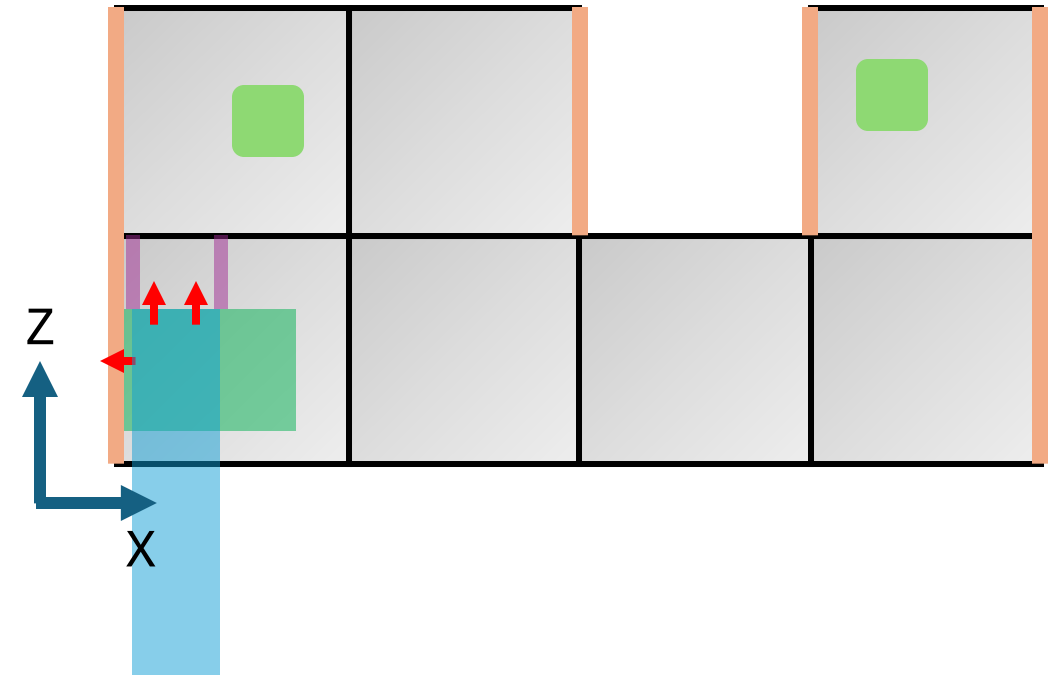
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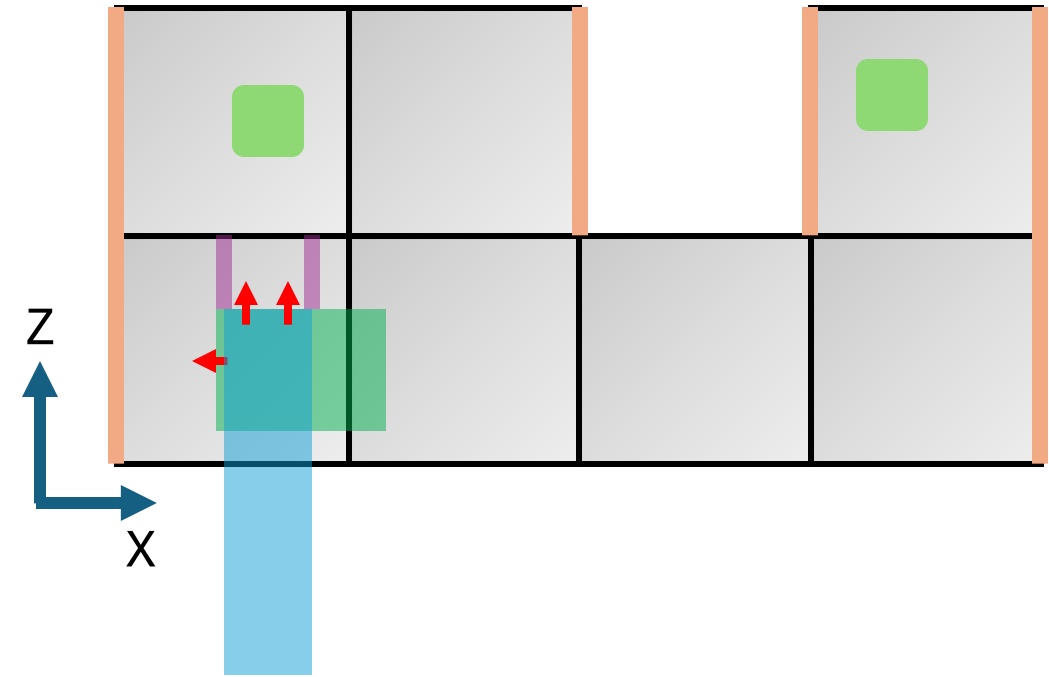


2. Drive to the position of the first „first-row-cube“ and grab it

2.1 Drive the system in position on the X axis using the X motor and the stored Cube positions

2.2 Extend the Grabber by moving it along the Z axis using the Z motor until the distance measured by the distance sensor is $<1\text{cm}$

2.3 Grab the Cube by using the Grabbing Arm controlled by the Grabbing Motor. Return the Grabber to $Z=0$ using the Z motor, and validate the success using the distance sensor (should still be $<2\text{cm}$).

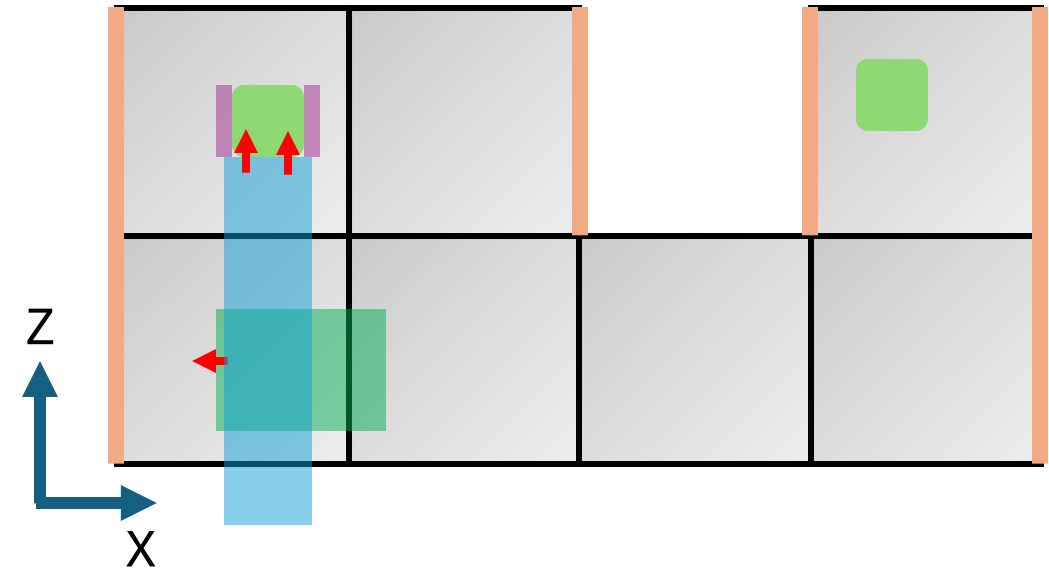


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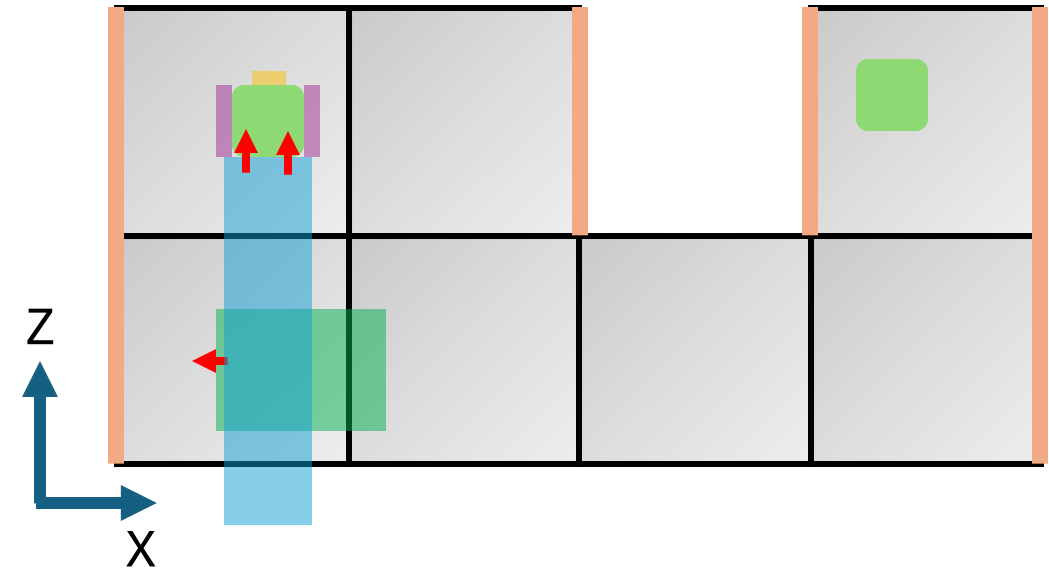


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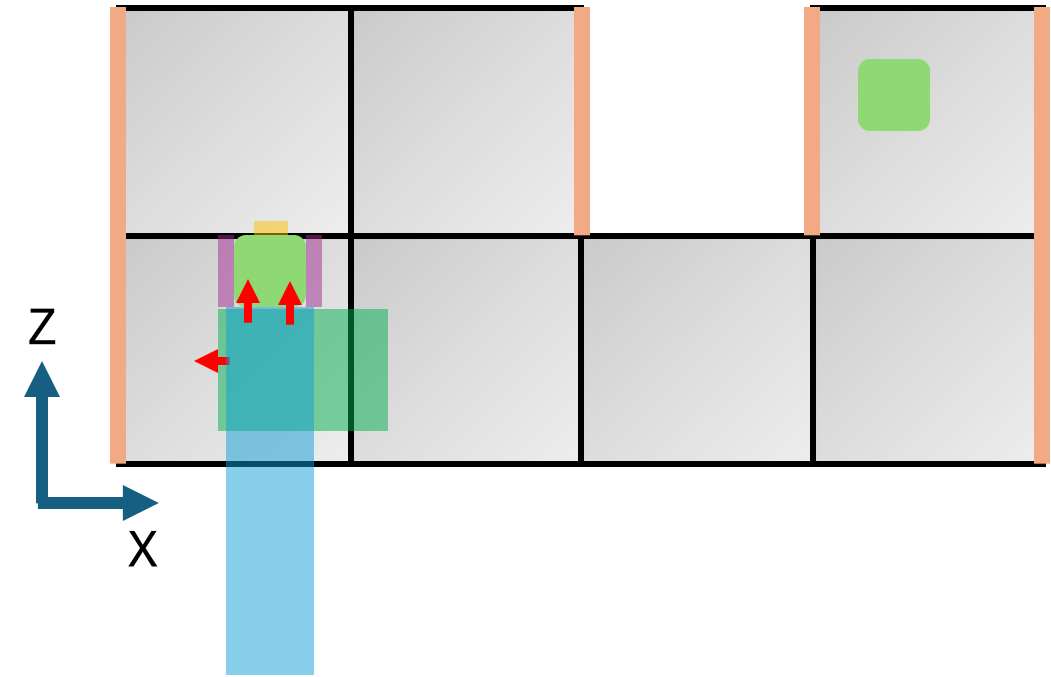
3. Determine the color of the Cube and sort it away

3.1 Determine the color of the Cube using the color sensor

3.2 Move the Cube to the corresponding pile position in X direction using the X motor

3.3 Push the cube outward to the specified depth in Z direction using the Z motor

3.4 Release the cube by Opening the Grabbing Arm with the Grabber Motor and Return to Z=0 with the Z-Motor



4. Drive to the position of the next „first row cube“

4.1 Repeat steps 2-4. If all „first row cubes“ are collected, repeat steps 1-4

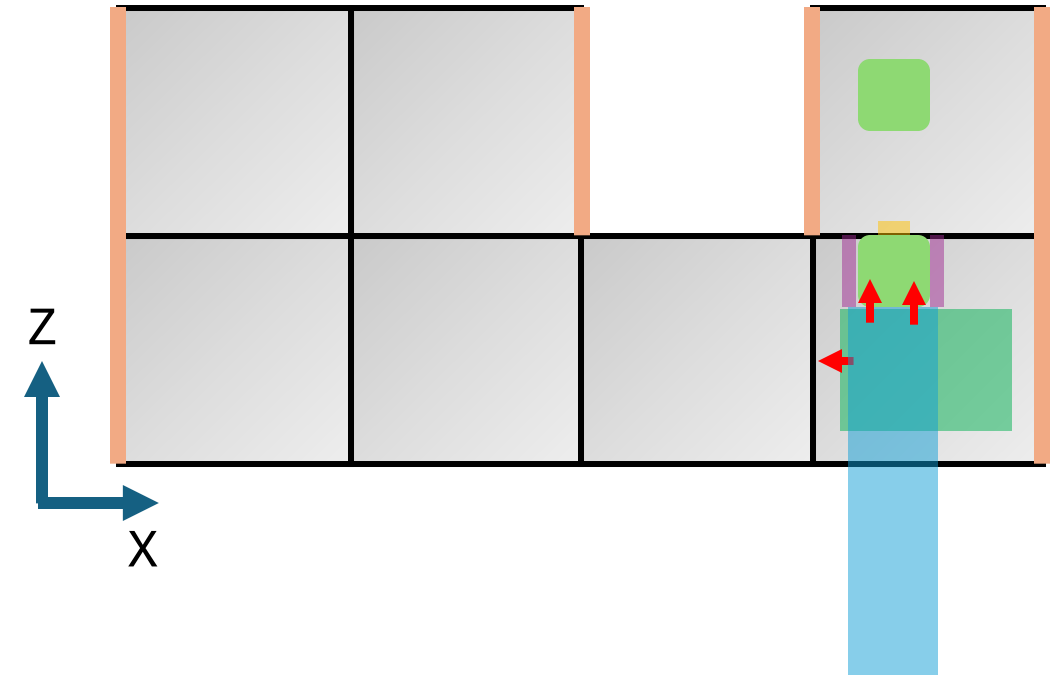
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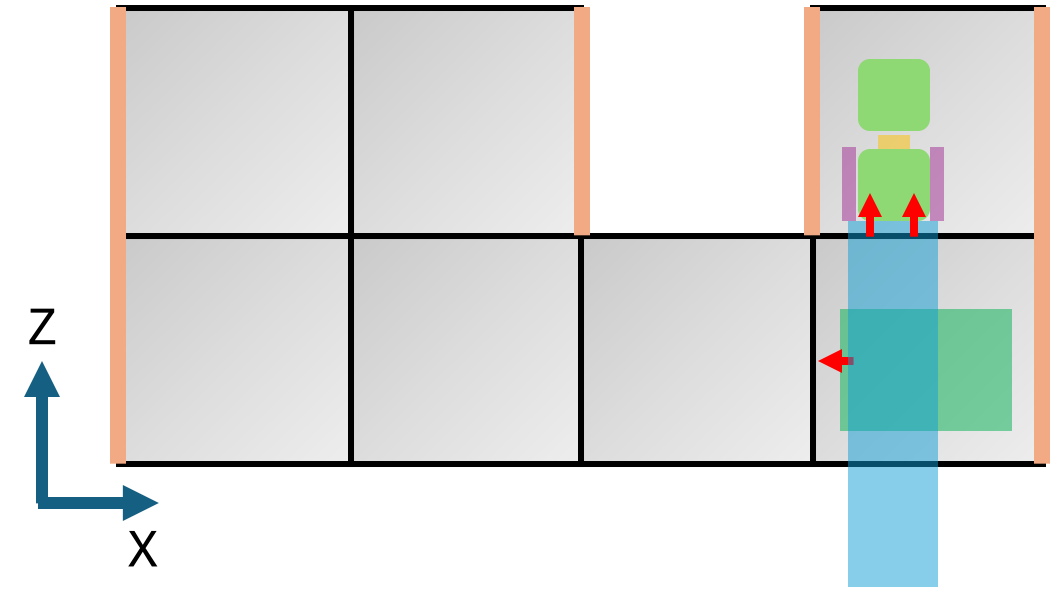
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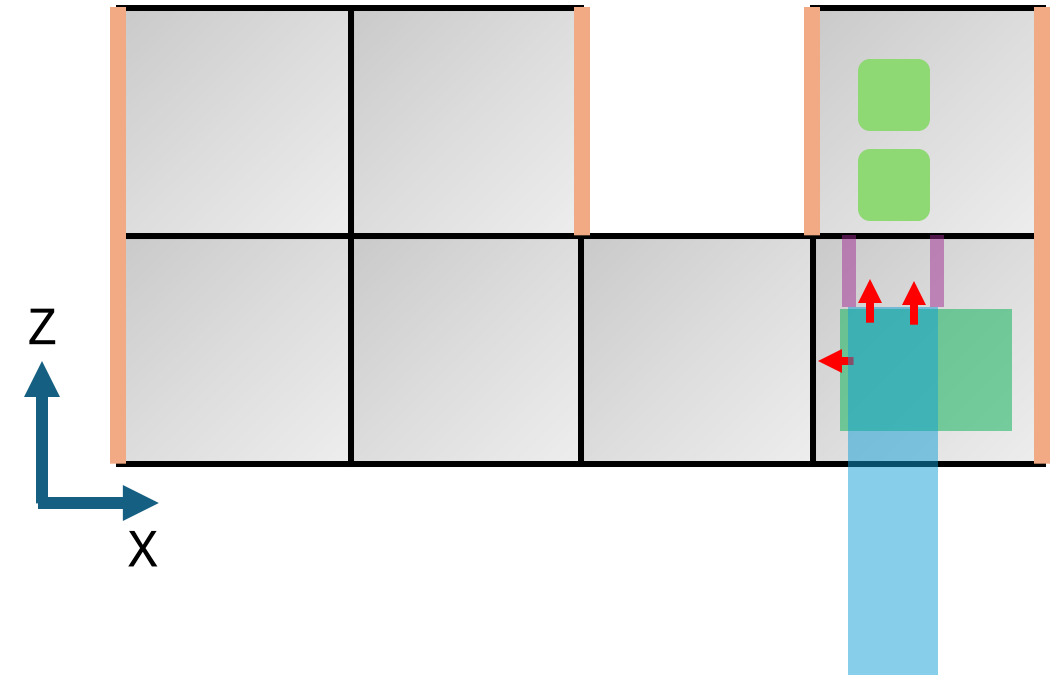
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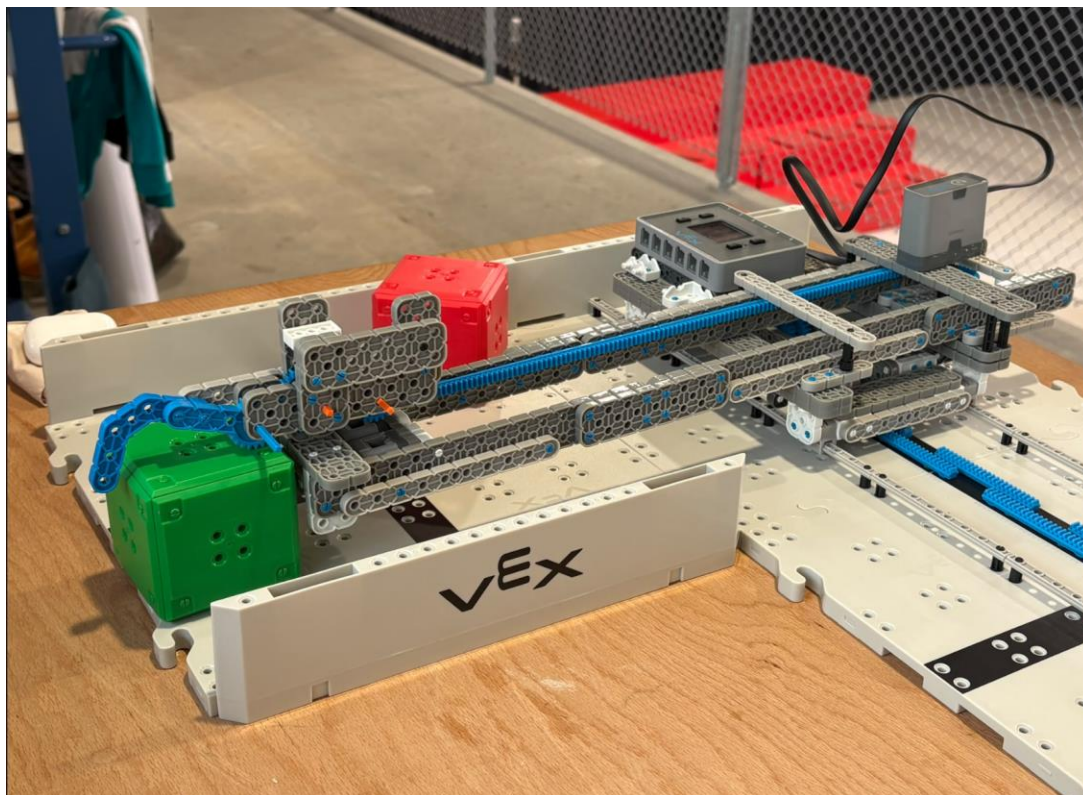
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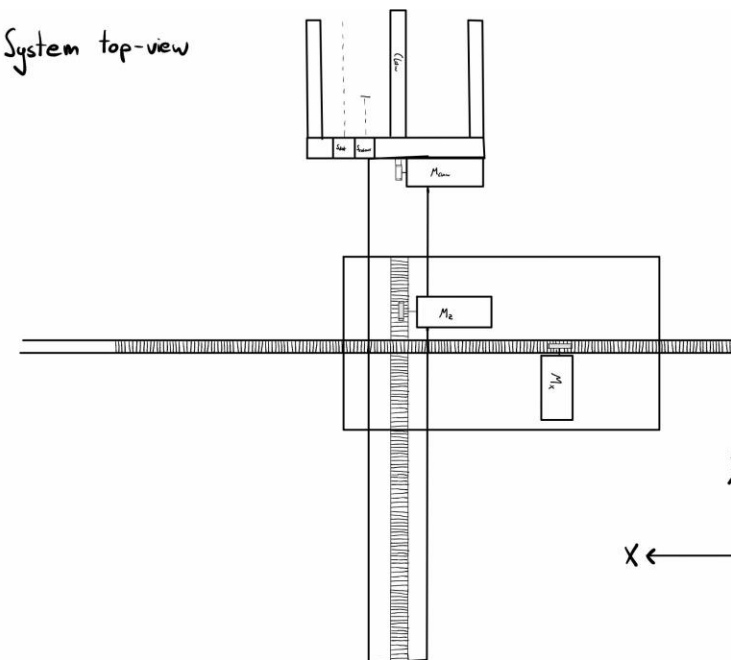
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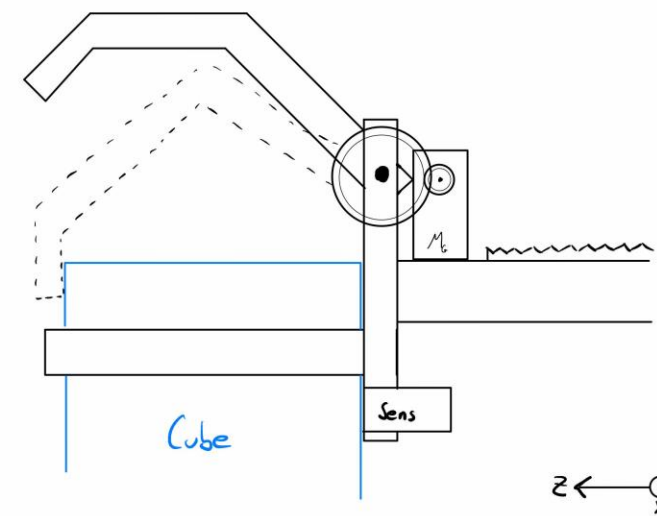
System Overview



System top-view



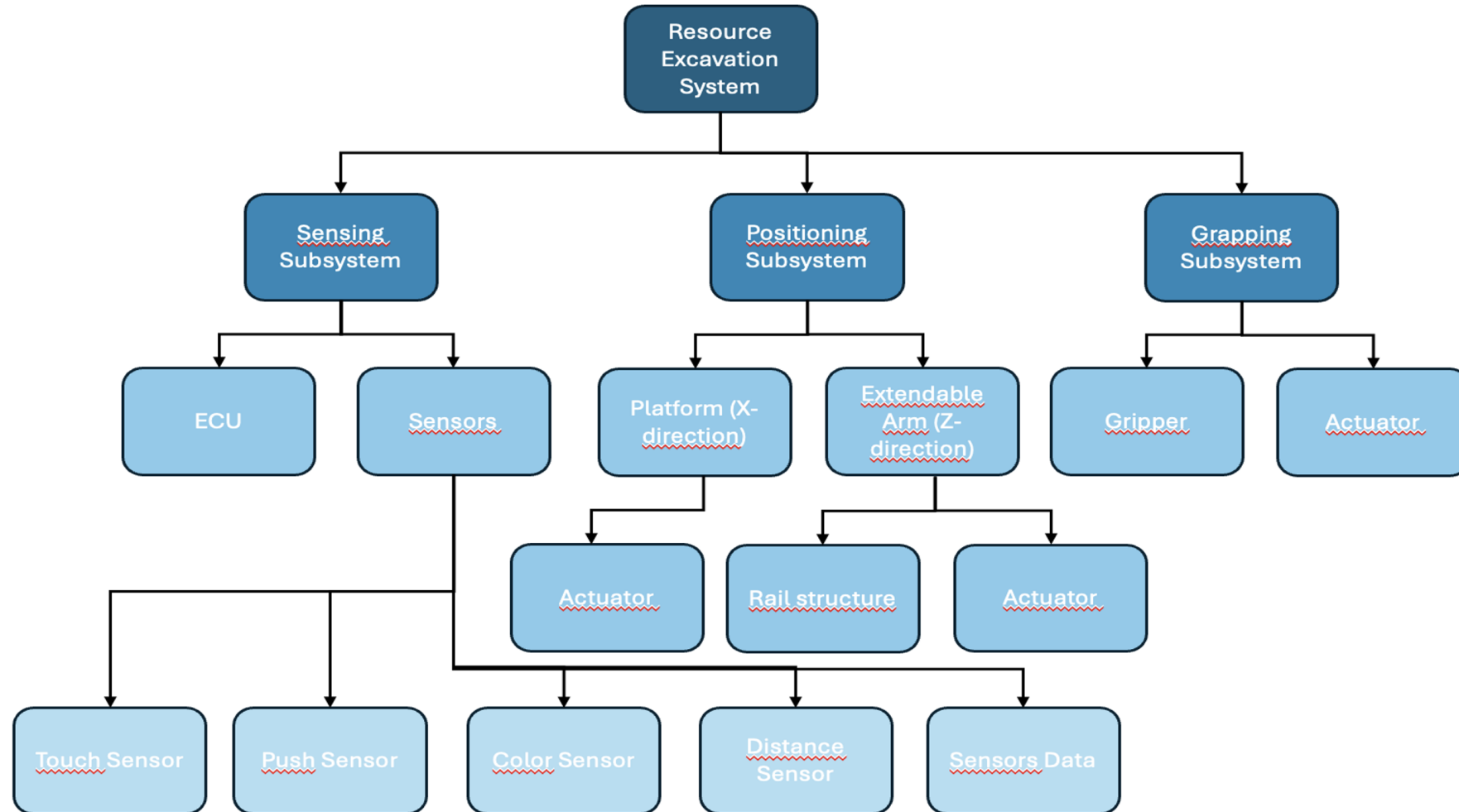
Gripper side-view



LO Requirement

Title	Description	Value
Resource Excavation System	The system shall collect cubes from the mining area, transport it to the production facility and sort them into piles autonomously	As many as possible (color preferences), only if within boundaries

Product Tree



V-Model

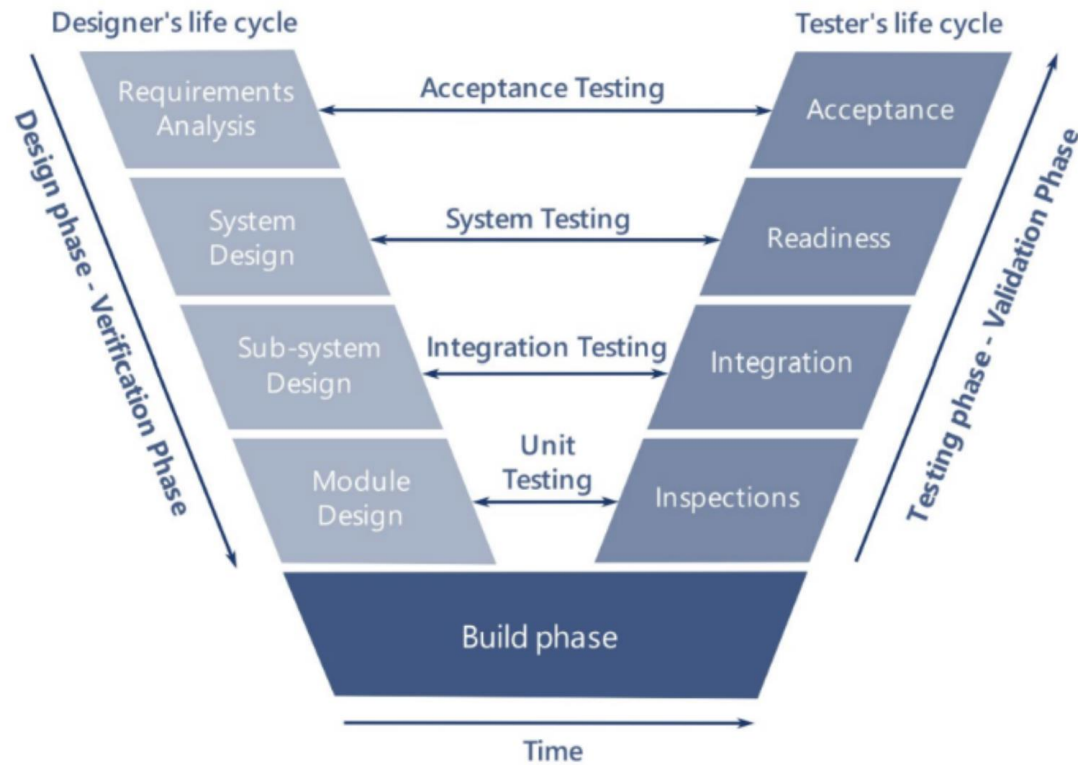
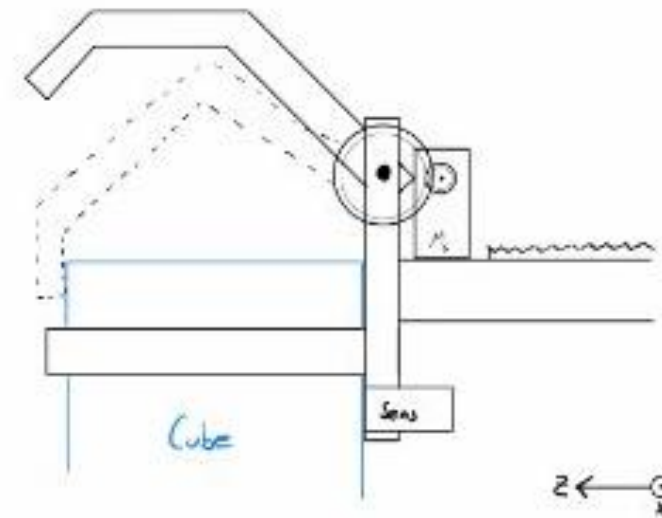


Figure 3: V-Model

- For needed autonomous system need sensing, positioning and gripping capability
- Looking at subsystem for gripping:

Gripper side-view



FMEA – Gripping Subsystem

Number	Structure	Function	Potential Failure	Potential Effect	Potential Reason	Risk Assessment				Elimination Measure	Assessment	Assignment
						O	S	D	RPN			
1)	Actuator	Drop arm upon sensor trigger	Arm already dropped or fails to drop	Cube cannot be positioned or pulled in (passes underneath)	Wrong sensor trigger (too early/none) / ECU signal loss	3	8	10	240	Track arm position (angle/inclination); add software timeout if sensor fails	Tracking & software timeout to be implemented; no additional contact sensor available	Team, validated by next Monday
2)	Actuator	Keep arm in down position	Arm lifted before destination reached	Loss of cube during transport	False sensor trigger / ECU signal noise	1	8	10 (3 with tracking)	80 (24)	Track arm position (angle/inclination) to catch unwanted movement immediately	Position tracking to be implemented	Team, validated by next Monday
3)	Actuator	Lift arm upon sensor trigger	Arm fails to lift at destination	No release of cube at production site	Missing sensor trigger / ECU signal failure	2	8	10	160	Track arm position (angle/inclination); implement software fallback release	Tracking & software timeout to be implemented; no additional distance sensor available	Team, validated by next Monday
4)	Actuator	Keep arm in position against force	Arm moved up against actuator force	Loss of cube during transport	High friction pulling arm open	1	8	10 (3 with tracking)	80 (24)	Track position to detect force displacement and actively compensate via PID controller	Test surface/holding friction ASAP; then implementation or otherwise failure accepted	Team, test ASAP
5)	Gripper Arm	Pull and hold cube in "hand"	Arm destroyed / snaps off	Loss of cube and full system loss (single arm failure)	Severe collision / High friction hazard during transport	1	9	10	90	Add second arm	Stress test mechanical durability. Add second arm if failure likely, if only caused by external hazard failure accepted	Team, test ASAP

FMEA – Gripping Subsystem

- **Single points of failure (SPOFs) in design:**
 - Single Actuator
 - Single Gripper Arm
- **Most critical failure mode:**
 - **#1: "Arm already dropped or fails to drop upon sensor trigger" (RPN = 240)**
 - **S = 8 (High):** Impossible to collect cube then
 - **O = 3 (Moderate):** Touch and distance sensors occasionally suffer from false triggers
 - **D = 10 (Maximum risk):** Without position tracking, ECU completely "blind" to this issue
- **Test:**
 - **Sensor & Actuator Response Timing:** Measure latency between sensor signal triggering and mechanical arm fully dropping
 - **Dynamic Drive & Trigger Validation (False-Trigger Test):** Verify the arm does not drop prematurely when driving past non-target obstacles
- **Mitigation measures:**
 - **Implement Position Tracking or Overturn arm**
 - **Software Timeout Fallback:** safety routine: if system in cube-collection phase and distance sensor confirms proximity to target cube software timeout forces arm to drop automatically even without trigger

A composite image showing a lunar base. In the foreground, there's a lunar rover with solar panels and a Chinese flag. In the middle ground, there are several white, modular structures and a lunar lander. In the background, the Earth is visible in the dark sky. The word "Thanks!" is overlaid in large white text.

Thanks!

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